

Problem 9

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Problem 1 Let $f(x) = (x - 3)^{-2}$. Show that there is no value c in $(1, 4)$ such that $f(4) - f(1) = f'(c)(4 - 1)$. Why does this not contradict the Mean Value Theorem?

Explanation. First notice that

$$f(4) - f(1) = 1^{-2} - (-2)^{-2} = 1 - \frac{1}{4} = \frac{3}{4}$$

and so we are looking for a value c such that

$$3f'(c) = \frac{3}{4} \implies f'(c) = \frac{1}{4}$$

Then since $f'(x) = \frac{-2}{(x - 3)^3}$, we can compute:

$$f'(c) = \frac{-2}{(c - 3)^3} := \frac{1}{4}$$

$$(c - 3)^3 = -8$$

$$c = 3 - 2 = 1$$

But 1 is not in the interval $(1, 4)$. This does not contradict the Mean Value Theorem since f is not continuous at $x = 3$.

